Progress Report

Weekly Project Update Overview

Project Development Timeline

Weeks 1-3 Sensing Architecture

Determine what sensing architecture to use as well as what vision and measurement to use on the robot.

Weeks 4–6 Mechanical & Actuation

Structural design of the 4-DOF robotic arm, choice of actuators, and the pneumatic gripper.

Weeks 7-9 Control & Programming

Build the control system with ROS2, STM32 motor controller and motion planning.

Weeks 10–12 Integrate all subsystems

Connect all subsystems and test the system on the smart pick-and-place operation.

Weekly Progress Goals

DONE

ONGOING

01

ROBOTIC ARM DESIGN FINALIZATION

The full CAD model of the robotic arm was finalized, including joint geometry, link dimensions, and end-effector integration. The design is now ready for simulation and control integration.

02

PNEUMATIC GRIPPER SYSTEM

A vacuum-based gripper was integrated into the end-effector. Suction force calculations were performed based on object weight, and the required pressure from the compressor was determined. Component selection is nearing completion.

03

MECHANICAL & CONTROL VALIDATION

Joint torque requirements were calculated based on arm geometry and payload. The full system model was converted into URDF and integrated into ROS2 MoveIt2 for motion planning and simulation using the actual robot geometry.

Weekly Timeline

Robotic arm CAD model was finalized, including joint geometry, link dimensions, and end-effector integration.

Vacuum gripper was integrated into the design and suction force calculations were performed to determine required compressor pressure.

Joint torque requirements were calculated based on arm geometry and payload, supporting motor sizing and actuator selection.

3

4

Full system model was converted into URDF and integrated into ROS2 MoveIt2 for motion planning and simulation with real geometry.

Goal # 1

Mechanical & Actuation Validation

Completed

PROGRESS CHECK

- Robotic arm CAD model was finalized, including joint geometry and link dimensions.
- Joint torque requirements were calculated based on arm configuration and payload.
- Vacuum gripper was integrated and suction force/pressure requirements were determined

BLOCKERS

- Verification of vacuum gripper performance under real operating conditions.
- Can’t test vacuum gripper in gazebo

ACTION ITEMS

- Validate gripper performance with physical testing or simulation.

Goal # 2

Pneumatic Gripper System

In progress

PROGRESS CHECK

- Vacuum gripper was integrated into the robotic arm end-effector design.
- Suction force calculations were performed based on object weight (ball/compass).
- Required pressure range from the compressor was determined for reliable gripping.

BLOCKERS

- Final selection of compressor and vacuum components pending availability and system constraints.
- Physical validation of suction performance has not yet been conducted

ACTION ITEMS

- Finalize vacuum pump/compressor selection based on required pressure and flow rate.

Goal # 3

Vision & Control system

In progress

PROGRESS CHECK

- Developed an initial ball tracking algorithm to detect and follow the target object, enabling autonomous positioning of the robotic arm.
- This tracking approach supports full 4-DOF operation by guiding the end-effector toward the object location.

BLOCKERS

- Real-time performance and accuracy of object tracking not yet validated on hardware.

ACTION ITEMS

- Complete ball tracking implementation and link it with ROS2 control.
- Integrate vision feedback with MoveIt2 for autonomous motion execution.
- Perform calibration and testing using the selected depth camera.

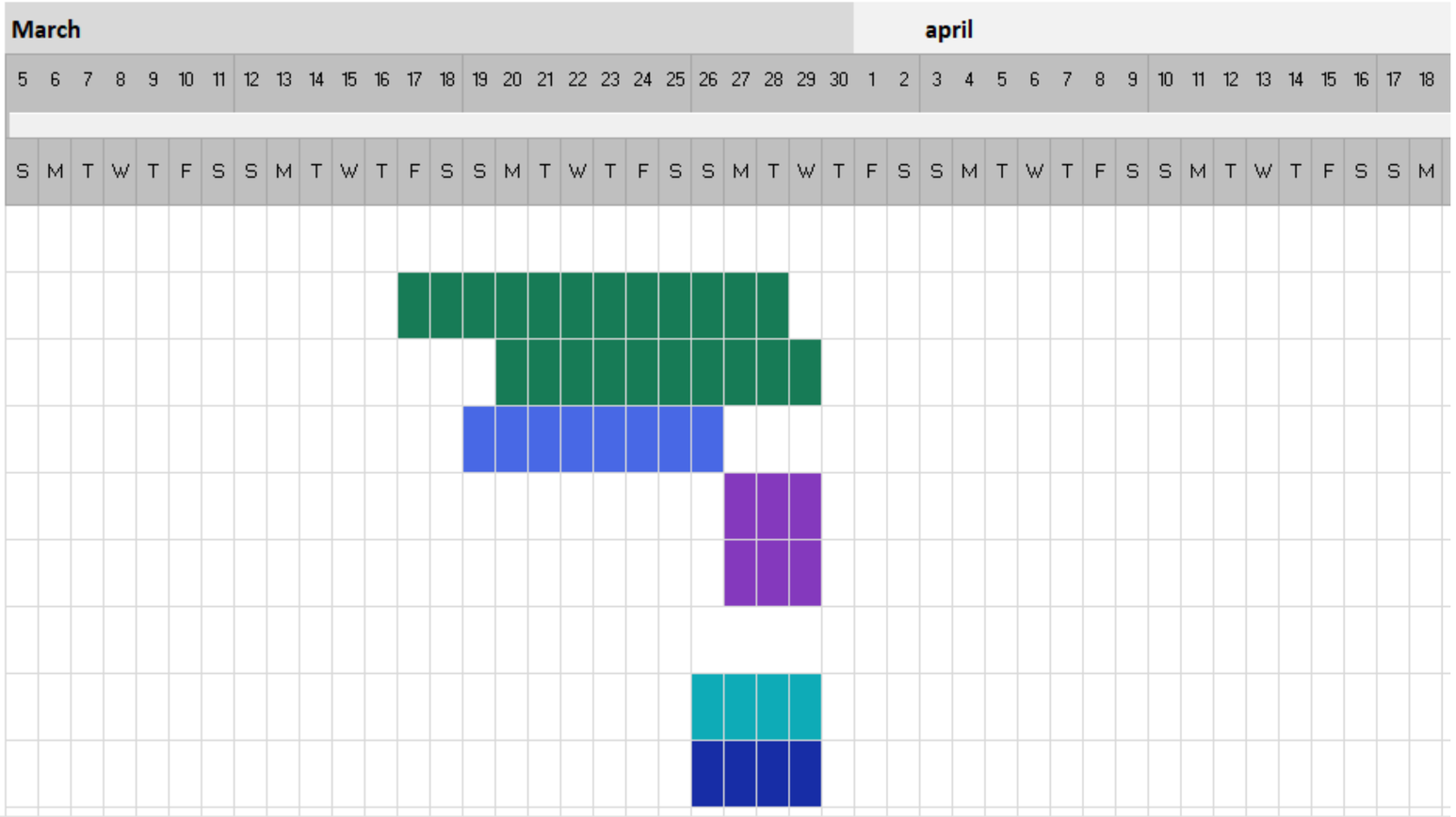
Project schedule (Gantt)

PROJECT: Robotic arm

Legend: On track | Low risk | Med risk | High risk | Unassigned

Scrolling increment: 5

Milestone description	Category	Assigned to	Progress	end	Days
Phase II: Mechanical & Actuation Finalization					
CAD model finalization of robotic arm	On Track	ɔdelaziz + (group effor	90%	3/31/2026	10
Vacuum gripper integration & suction force calculations	On Track	Hamza	85%	4/5/2026	7
Joint torque calculations based on arm geometry	Low Risk	Khaled	80%	3/28/2026	10
Ideamaker simulation & print parameter optimization for ioints	Low Risk	Abdelaiz	50%	4/20/2026	3
Ideamaker simulation & print parameter optimization for gripper	Low Risk	Hamza	50%	4/6/2026	3
Phase III: Control & System Integration					
URDF conversion from STEP model	Low Risk	omar	80%	4/6/2026	4
openCV ball tracker	Low Risk	omar	40%	4/8/2026	4



Omar’s progress

Time frame:

Week 4

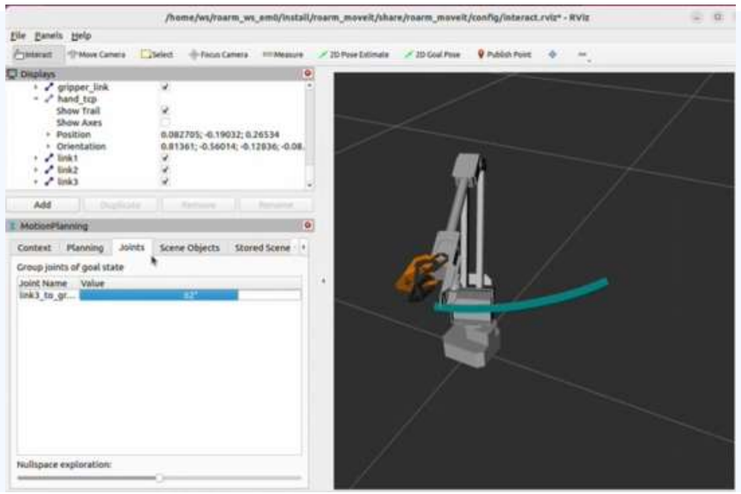
Main role:

Software & Ros2

Accomplished (since last report)

- Implemented final URDF of the model in gazebo and Rviz2 for moveit2 control based on final geometrical features
- Made an OpenCV classifier that follows a red ball for autonomous vehicle for 2D tracking as of now

Evidence (show, don’t tell)



Next Week Plan

- Use a model for object detection and tracking of a red ball and connect a depth camera and see how it behaves

In Progress

- Implement ros2_control with the robotic arm to achieve full autonomy

Help Needed / Questions

- Hardware specs confirmation
- Motor driver compatibility
- ROS2 integration issues

Hamza’s progress

Time frame:

Week 4

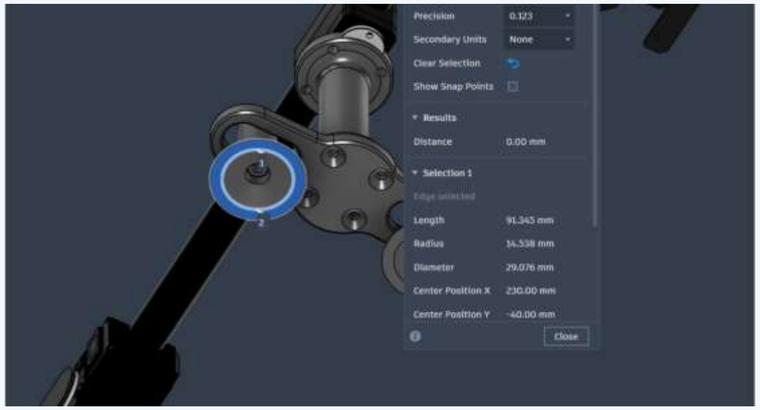
Main role:

Gripper(pneumatic)

Accomplished (since last report)

•Refined pneumatic gripper design (double-acting cylinder).
•Developed CAD model and defined gripping dimensions.
•Estimated required gripping force based on object weight.
•Calculated required operating pressure using piston area.

Evidence (show, don’t tell)



In Progress

•Finalizing cylinder specifications (bore & stroke).
•Refining CAD design and mounting interface.
•Preparing component list for procurement.

Next Week Plan

•Complete final CAD design.
•Start hardware prototyping.
•Implement and test valve control with Arduino.
•Validate gripping performance experimentally.

Help Needed / Questions

•Compressor specifications and pressure limits.
•Recommended operating pressure range.
•Gripper mounting compatibility with robotic arm

Khaled's progress

Time frame:

Week 4

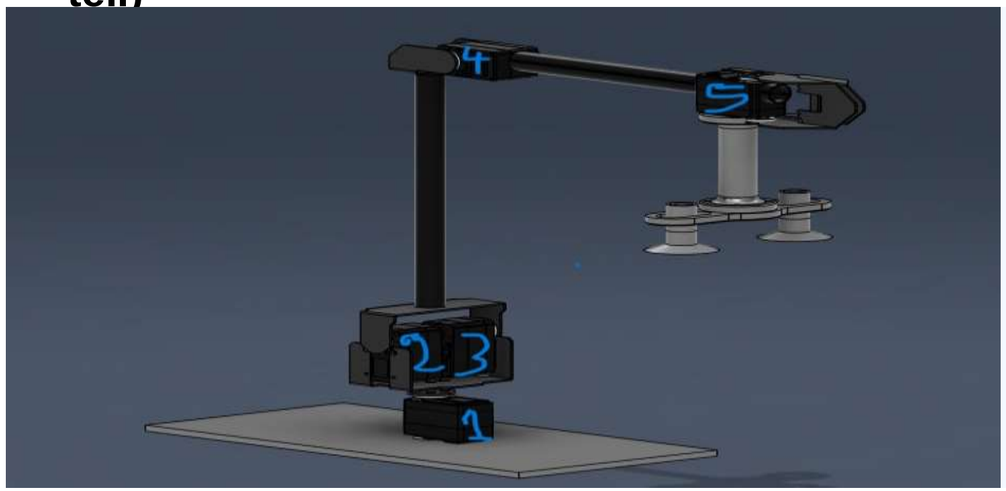
Main role:

Actuator Selection& Torque Calculation

Accomplished (since last report)

- Defined a 5-DOF actuator configuration, where each actuator controls motion in a different axis.
- Combined Actuator 2 and 3 to increase force and improve the ability to lift heavier loads.
- Selected a high-torque servo motor with encoder for accurate and reliable control.
- Performed detailed torque calculations for each actuator based on arm dimensions and expected loads.

Evidence (show, don't tell)



Next Week Plan

•Finalize actuator specifications based on torque requirements

•Choose the appropriate motor driver for the chosen actuator(s)

•Begin hardware integration planning

In Progress

- Calculating the required torque for each actuator.
- Researching the integration of actuators with the Arduino/ROS2 control systems.

Help Needed /

•Confirmation of maximum payload weight for accurate calculations

•Verification of safety factor assumptions

Abdelaziz’s progress

Time frame:

Week 4

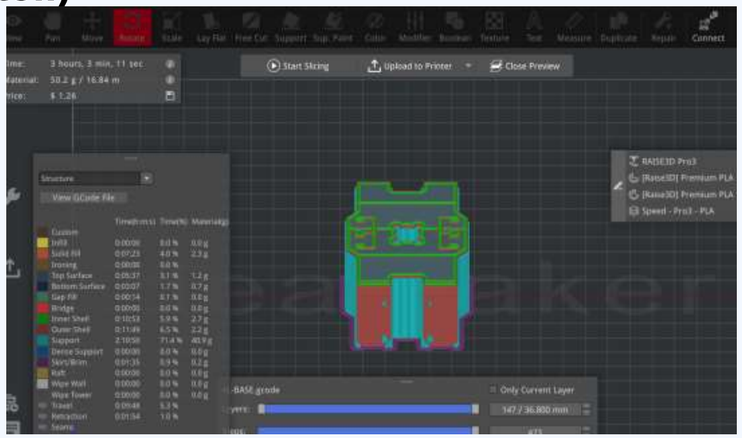
Main role:

Mechanical structure

Accomplished (since last report)

•Rebuilt the robotic arm CAD model with accurate dimensions
•Redesigned joints and connections for improved rigidity
•Selected optimal print settings (supports, infill, shell thickness)
•Verified printability using slicing preview

Evidence (show, don’t tell)



Next Week Plan

•Finalize all mechanical designs for full assembly
•Begin 3D printing key components
•Test fit and alignment of printed parts
•Perform basic strength and performance tests

In Progress

- Optimizing support structures for print quality and removal
- Fine-tuning infill density and shell thickness
- Evaluating joint performance and motion
- Preparing final files for printing and assembly

Help Needed / Questions

- How to improve joint durability and load capacity
- Feedback on selected print settings
- Ways to enhance overall mechanical stability
- Recommendations for stronger printing materials